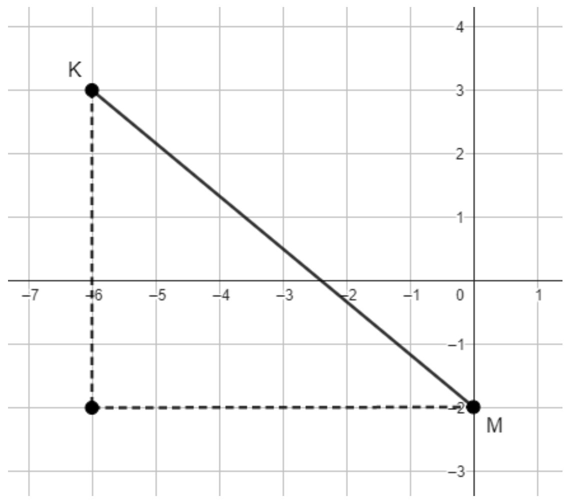
Geometry
Unit 13
Lesson 6 Homework

Name___Answer Key_______
Date______________________
Period_____________________

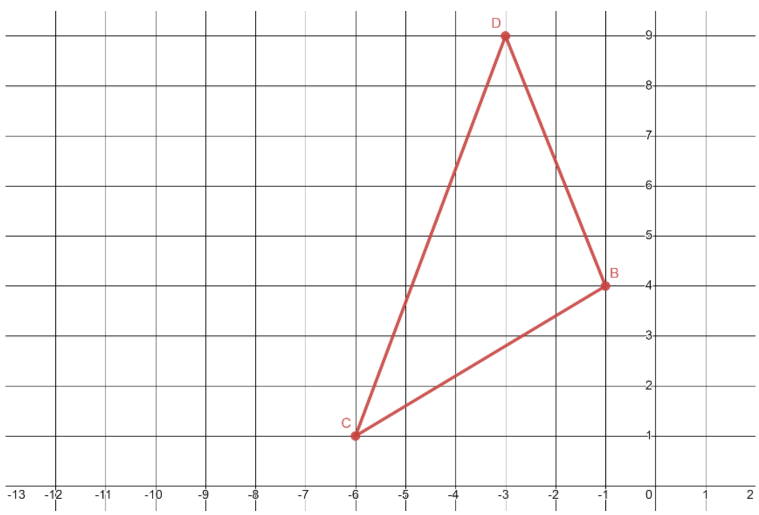
1. Find the length of $\overline{KM}$ using the Pythagorean Theorem. If necessary, round to the nearest thousandth.

$\sqrt{6^2+5^2}=\sqrt{61}\approx 7.810$



Use the following information for questions 2-4. $\triangle BCD$ has vertices at $B(-1,4)$, $C(-6,1)$, and $D(-3,9)$.

2. Graph $\triangle BCD$ on the coordinate plane.



3. Find the length of each segment. If necessary, round to the nearest tenth.

Line Segment	Length of Line Segment
$\overline{BC}$	$\sqrt{3^2+5^2}=\sqrt{34}\approx 5.8$
$\overline{CD}$	$\sqrt{3^2+8^2}=\sqrt{73}\approx 8.5$
$\overline{BD}$	$\sqrt{2^2+5^2}=\sqrt{29}\approx 5.4$

4. Complete the statement.

$\triangle BCD$ is classified as a(n)

- o equilateral
- o isosceles
- ✖ scalene

triangle because $\triangle BCD$ has

- ✖ no
- o two
- o three

equal side lengths.

Use the following information for questions 5-6. $\triangle FGH$ has vertices at $F(-2,0)$, $G(2,5)$, $H(7,1)$.

5. Find the length of each segment. If necessary, round to the nearest tenth.

Line Segment	**Length of Line Segment**
$\overline{FG}$	$\sqrt{4^2+5^2}=\sqrt{41}\approx 6.4$
$\overline{GH}$	$\sqrt{5^2+4^2}=\sqrt{41}\approx 6.4$
$\overline{FH}$	$\sqrt{9^2+1^2}=\sqrt{82}\approx 9.1$

6. Complete the statement.

$\triangle FGH$ is classified as a(n)

- o equilateral
- ✖ isosceles
- o scalene

triangle because $\triangle FGH$ has

- o no
- ✖ two
- o three

equal side lengths.